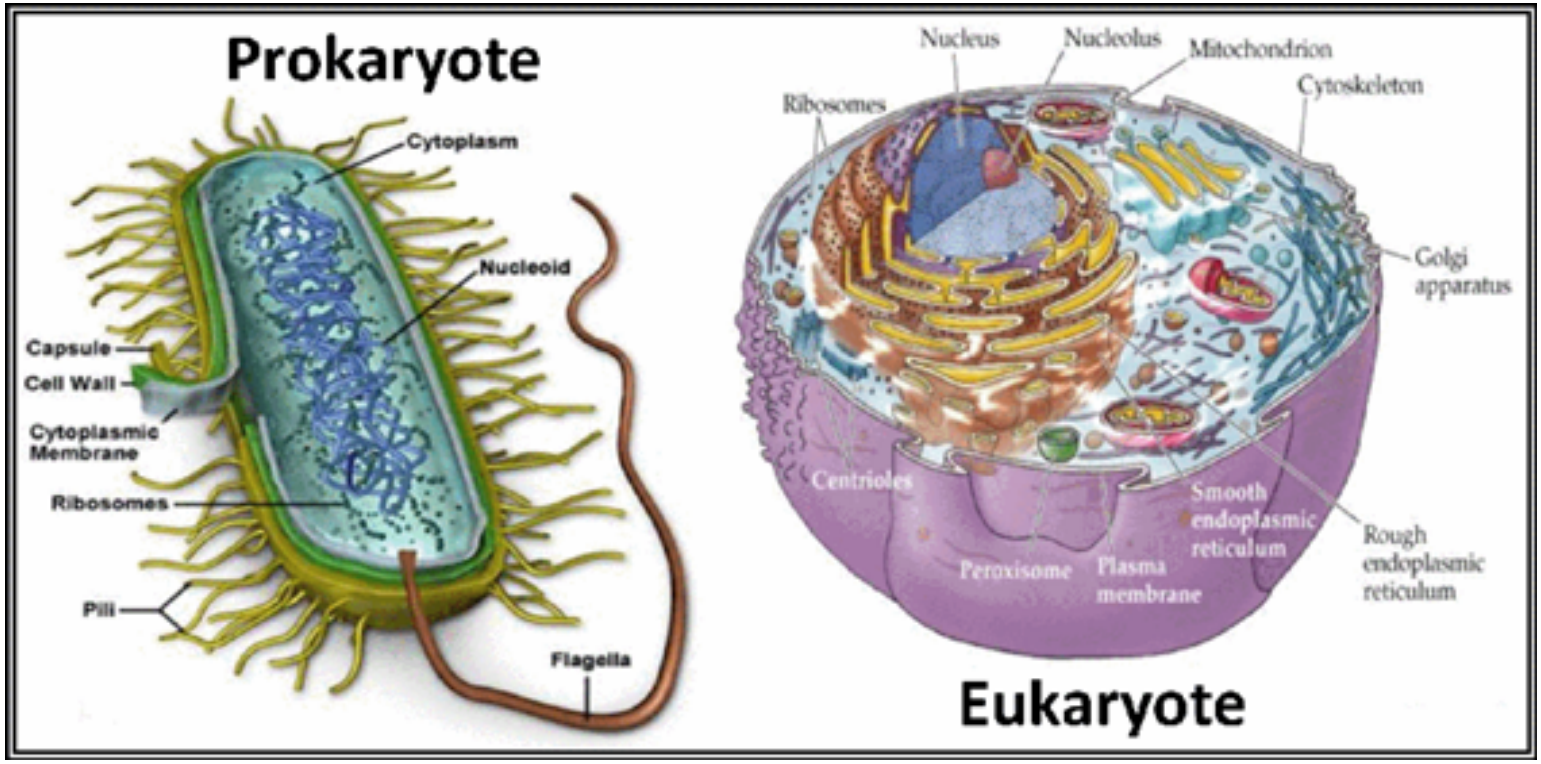
Computational Biology

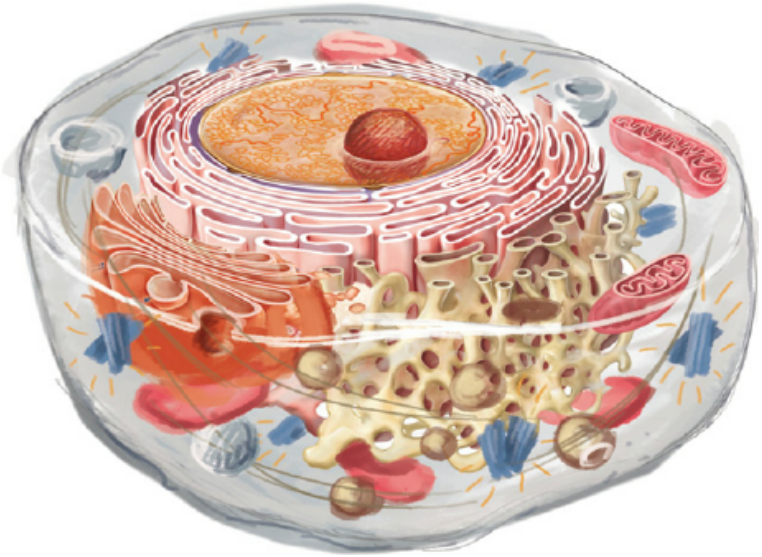
Lecture 2: Cell Regulation



Gene Regulation Networks

Outline

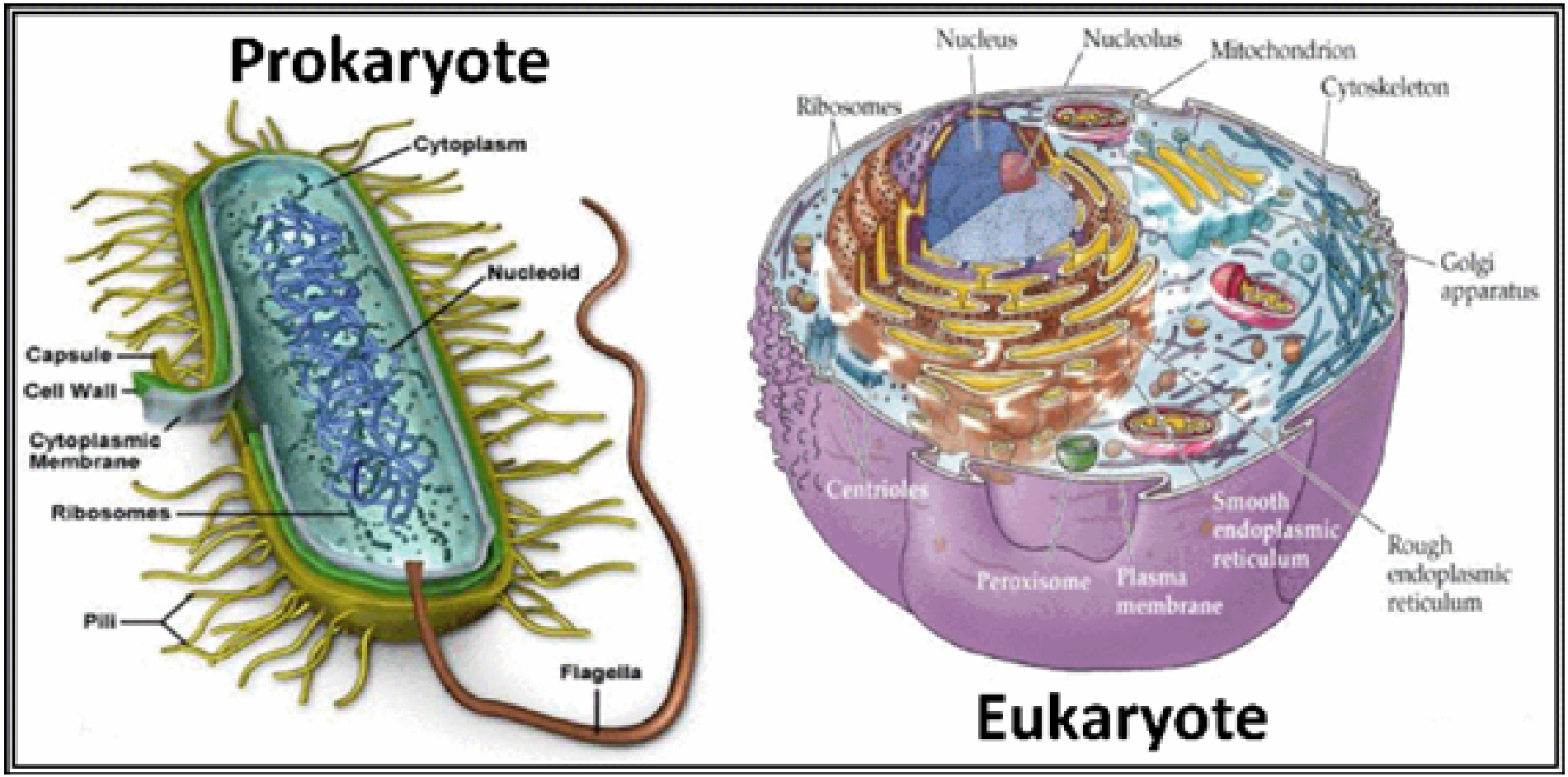
1. **Cells**
2. Protein Function
3. Cell Regulation



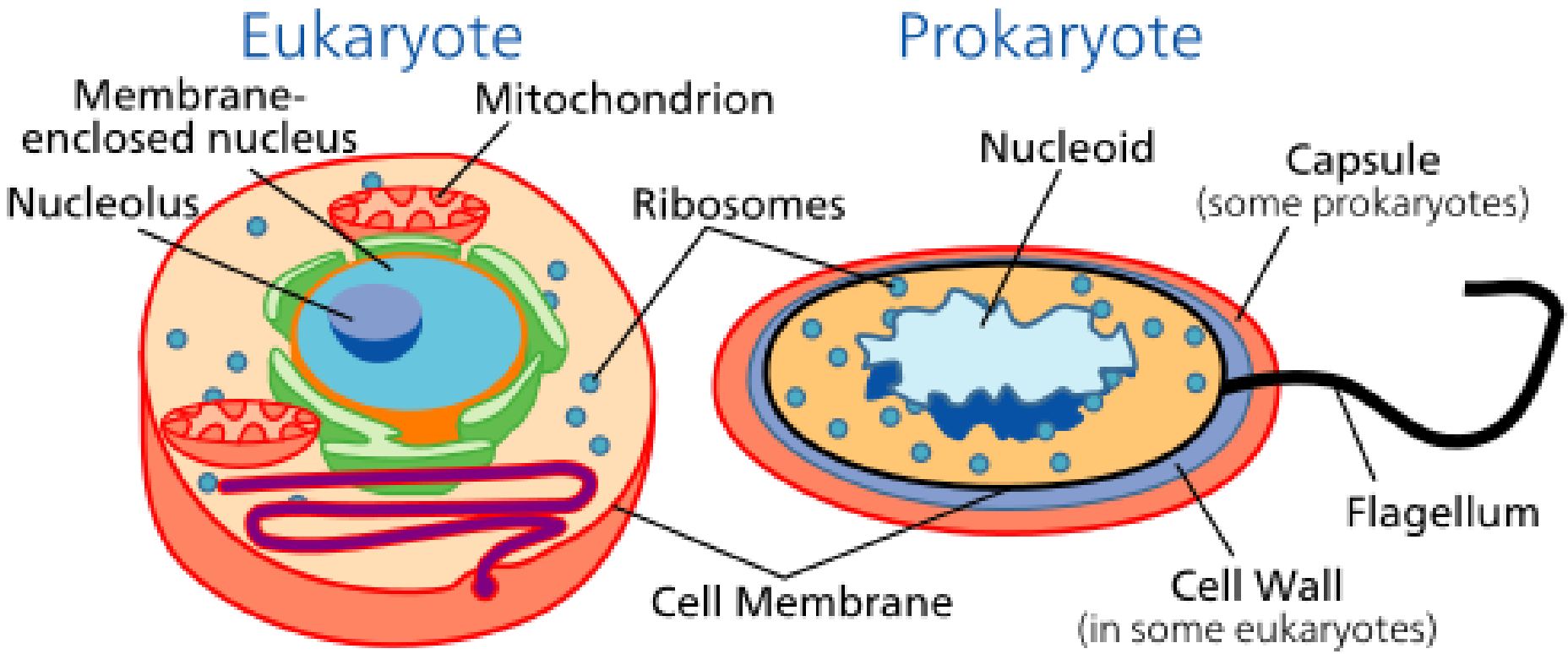
Cells

- Cells are complicated machines
- Understanding how they work is still an on-going story
- We are going to concentrated on parts where computational biology has and is making an impact

Prokaryotes and Eukaryotes



Prokaryotes and Eukaryotes



Differences and Similarities

- Share cell membrane, DNA code, ribosome (machine for producing proteins)
- Eukaryotes are much more complex
- It is now generally accepted that Eukaryotes were formed by fusing different cells
  - ⋆ nucleus
  - ⋆ mitrochondria (powerhouse, inherited from mother only)
- Eukaryotes can be multicellular

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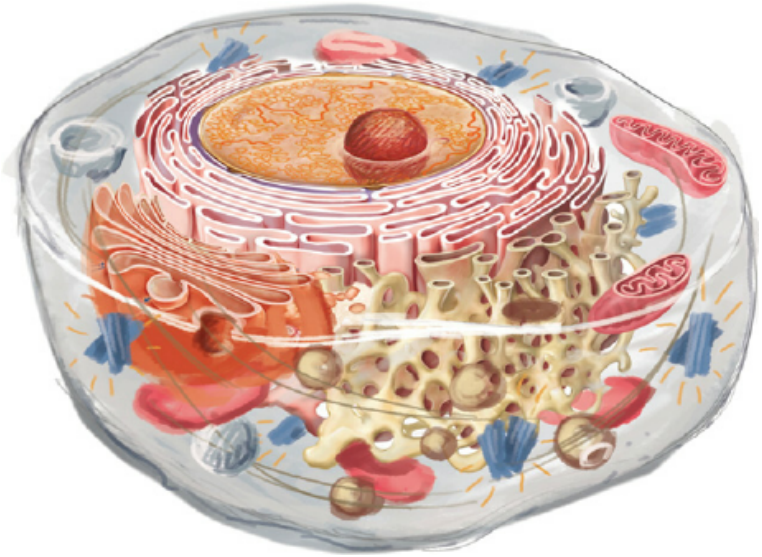


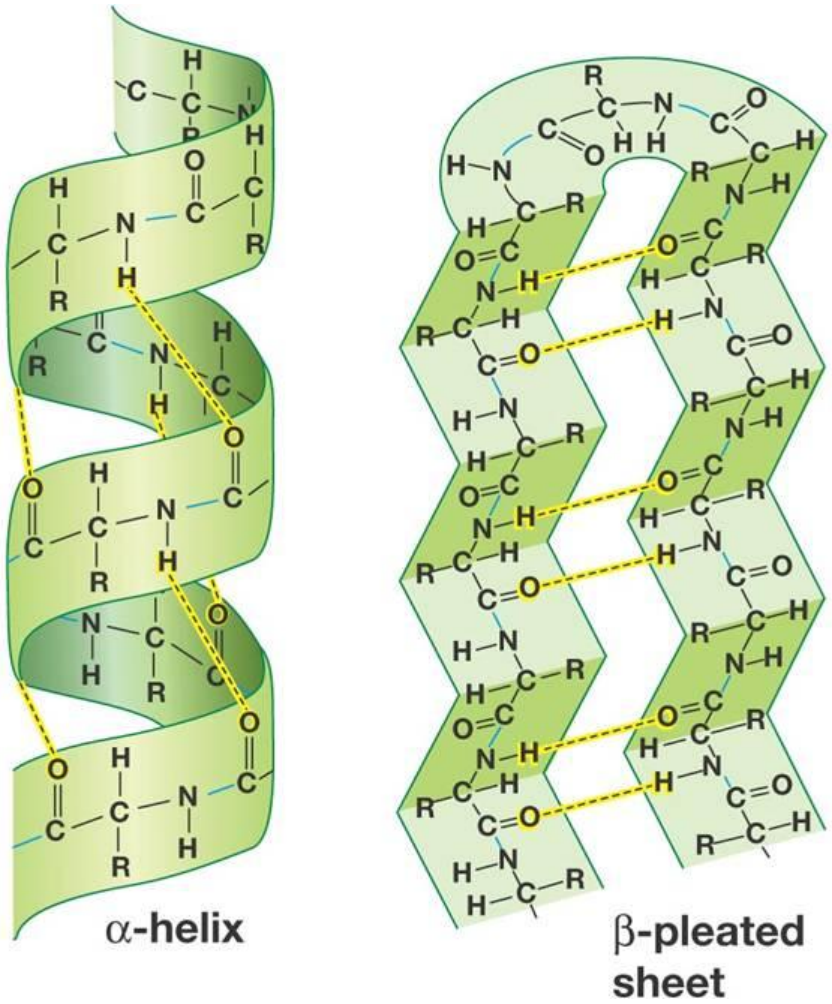
Understanding Proteins

- Proteins are hard to understand
- They are hard to sequence directly
- Fortunately we can deduce their sequence from DNA or mRNA
- Measuring protein concentration is difficult
  - ⋆ field of **proteomics** looks at this
  - ⋆ uses tools like mass spectrometer to count proteins
  - ⋆ beginning to do this much more often

Protein Folding

- To understand what proteins do we want to understand their three dimensional shape
- Alas this is hugely difficult
- In some instances we can crystallise them and use X-ray diffraction
- Predicting the structure is a non-local problem
- Rather dramatically, this problem has almost been solved by alpha-fold

Secondary Structure

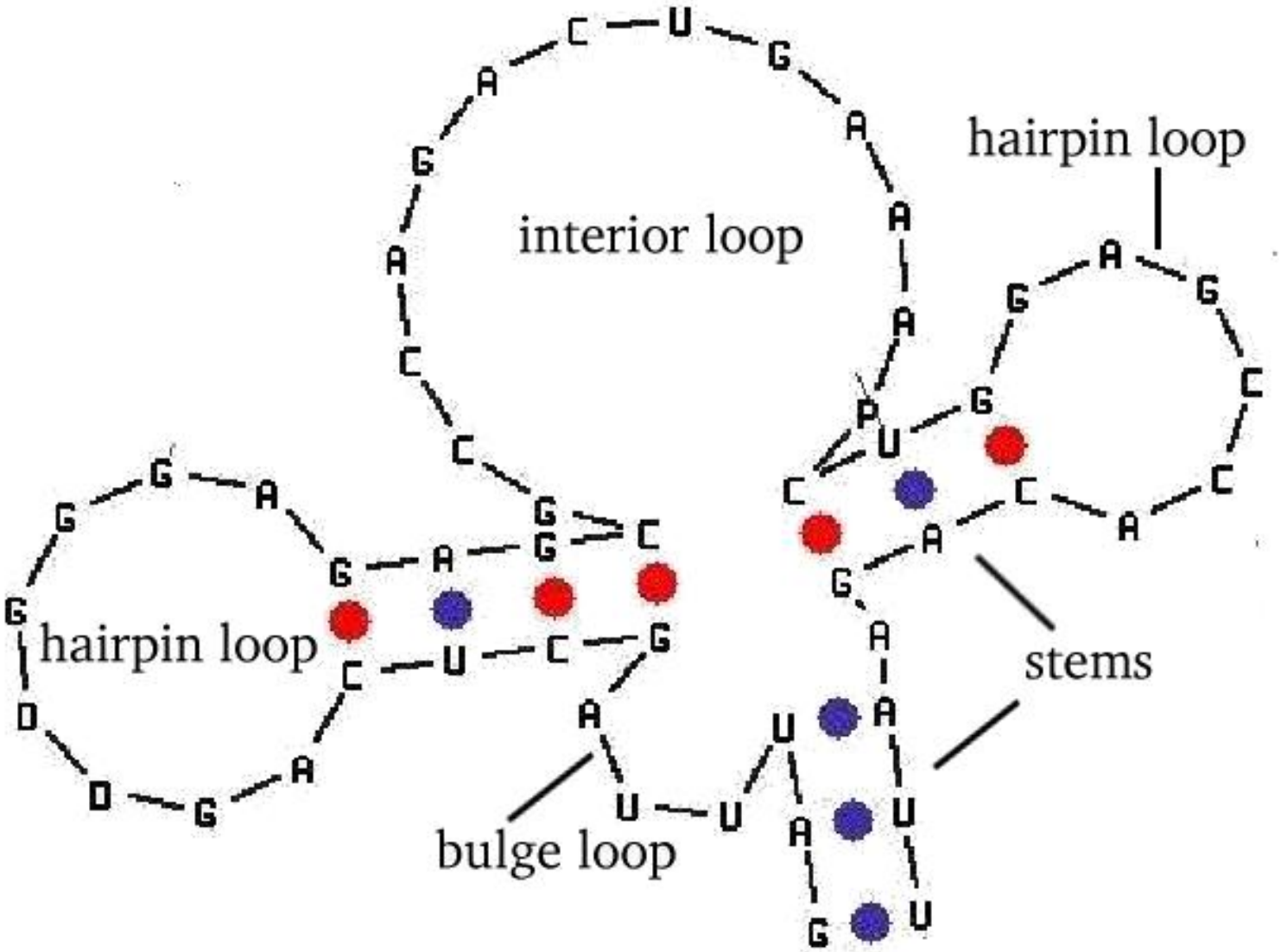


- The predominant components making up proteins are
  - ⋆ Alpha helices
  - ⋆ Beta sheets
  - ⋆ Various folds and connecting parts
- This is known as the secondary structure
- We will look at secondary structure prediction later in the course

Folding

- Protein folding is hard because typically amino acids some distance apart will interact through hydrogen bonds
- RNA also folds (it has a much more flexible backbone than DNA)
- RNA folding is considered much easier to solve
- Look for matching sub-sequences

RNA Folding



The First Law of Protein Sequences

Similar structures perform similar functions

- We can learn about the function of a protein by comparing with known proteins
- Because of evolution, many proteins are common to a huge number of organism
- Caveat: the converse is not necessarily true; dissimilar structures can have a similar function (convergent evolution)

Sequence Alignment

Mouse *Pax6* gene:

GTATCCAACGGTTGTGTGAGTAAAATTCTGGGCAGGTATTACGAGACTGGCTCCATCAGA

Fly *eyeless* gene:

Genetic similarity to mouse: 76.66%
Protein similarity to mouse: 100%

GTATCAAATGGATGTGTGAGCAAAATTCTCGGGAGGTATTATGAAACAGGAAGCATACGA

Shark eye control gene:

Genetic similarity to mouse: 85%
Protein similarity to mouse: 100%

GTGTCCAACGGTTGTGTCAGTAAAATCCTGGGCAGATACTATGAAACAGGATCCATCAGA

Squid eye control gene:

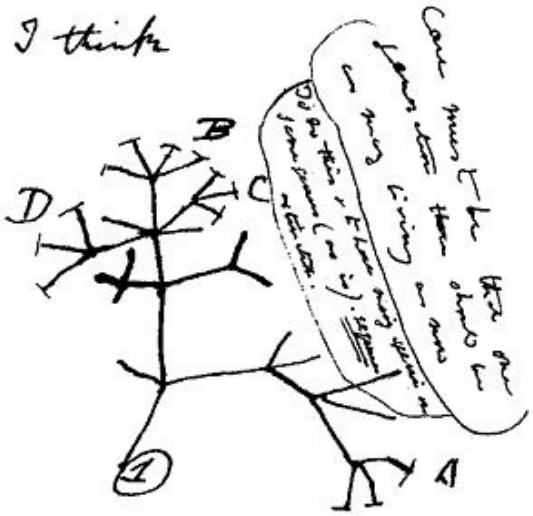
Genetic similarity to mouse: 78.33%
Protein similarity to mouse: 100%

GTCTCCAACGGCTGCGTTAGCAAGATTCTCGGACGGTACTATGAGACGGGCTCCATAAGA

Flatworm eye control gene:

Genetic similarity to mouse: 71.66%
Protein similarity to mouse: 100%

GTGTCTAATGGTTGTGTTAGTAAAATACTTTGCCGATATTATGGAACAGGTTCTATTAAA



Finding Aligned Sequences

- Biologists will routinely compare any new sequence with sequences already found
- This requires comparing with very large databases
- Typically sequences contain annotations indicating what someone believes a protein does in some organism
- We will spend sometime looking at alignment algorithms

Outline

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2. Protein Function
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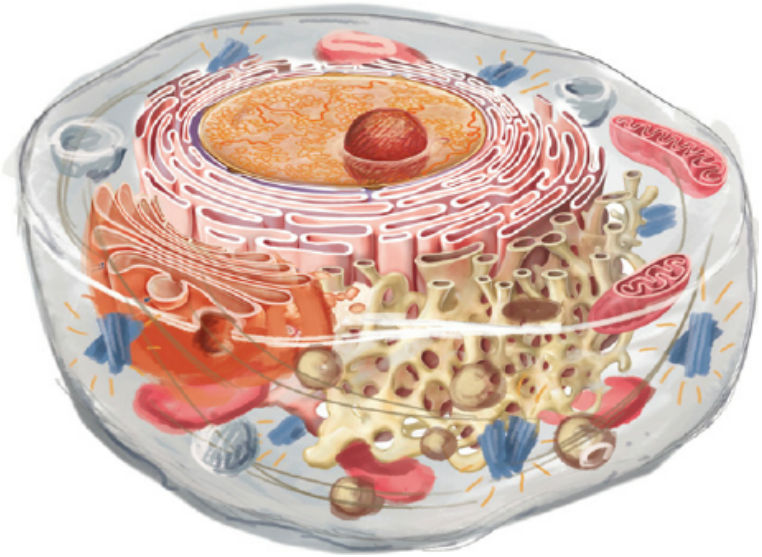
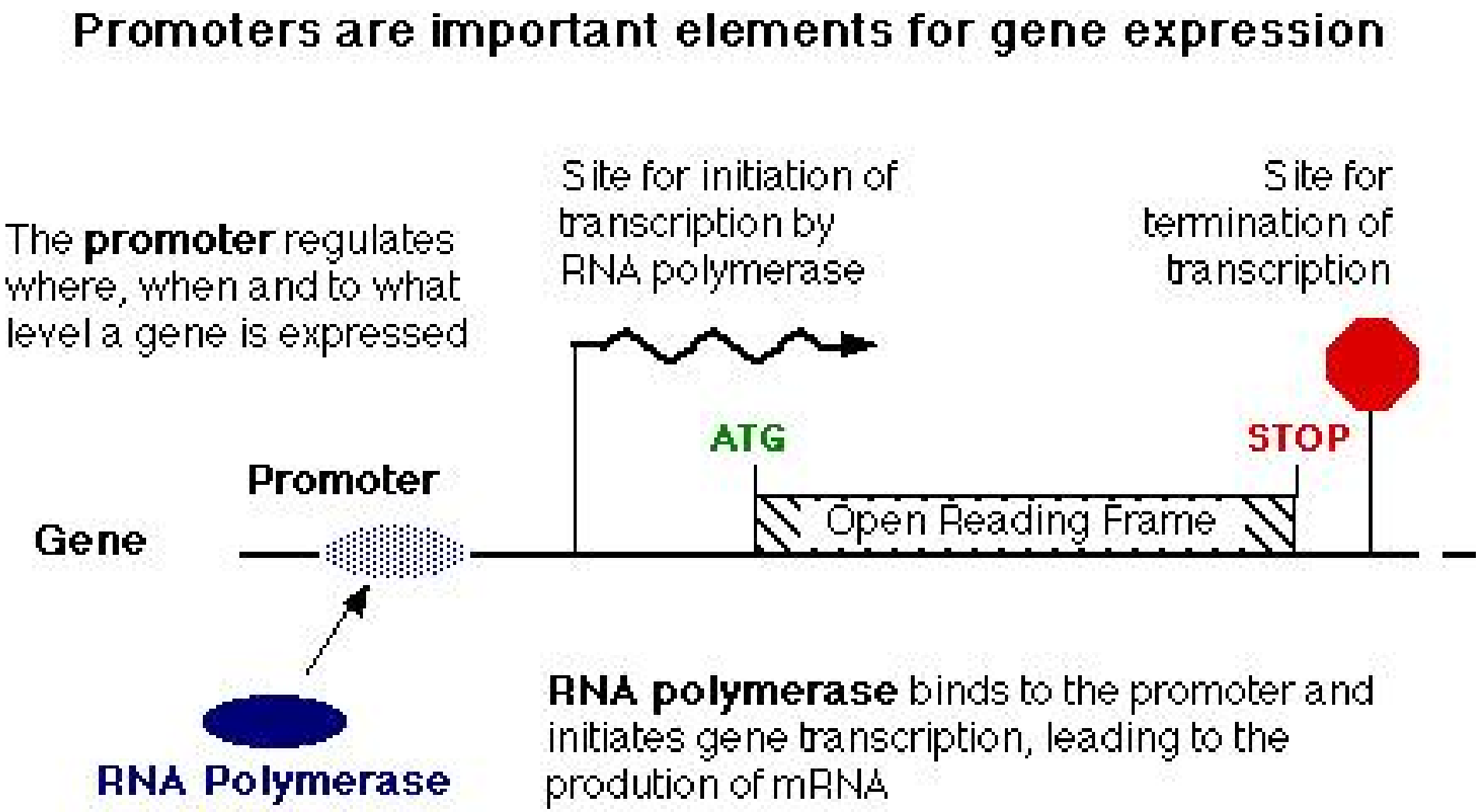


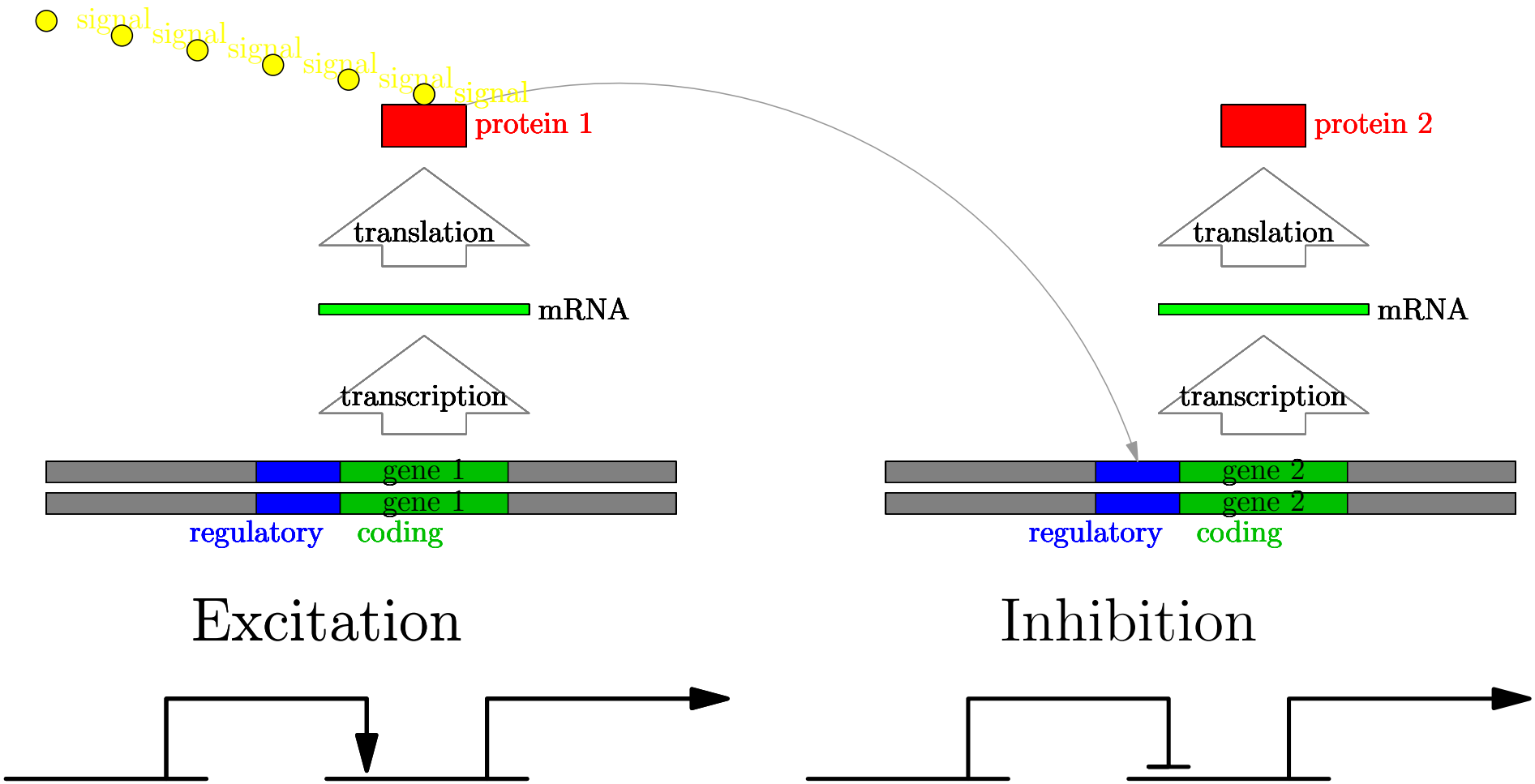
Cells Do

- Cells must respond to their environment, maintain their environment, divide
- They therefore need a control mechanism
- We are at the beginning of understand these mechanisms
- Proteins and DNA lie at the heart of this

Promoter Region



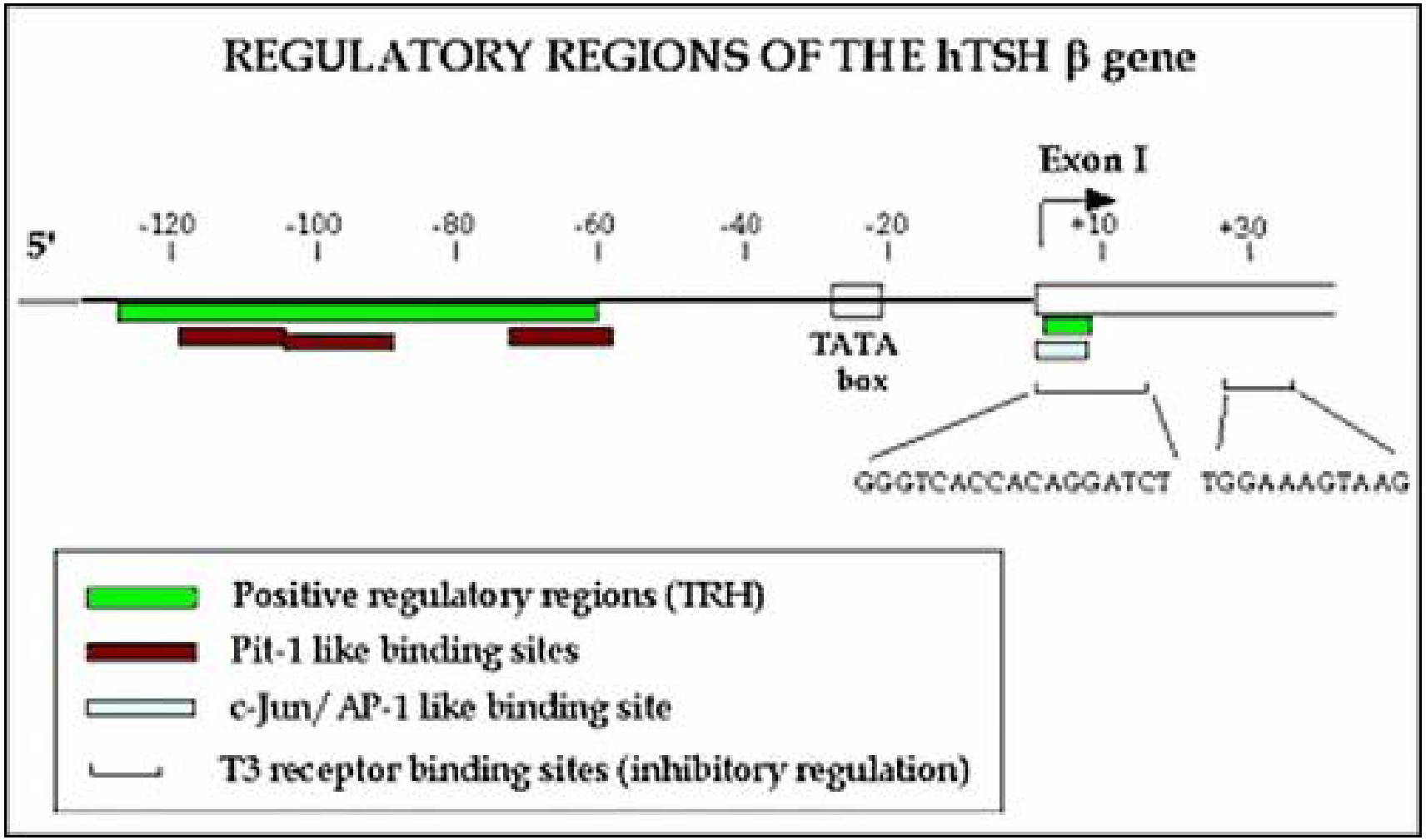
Regulation by Signal



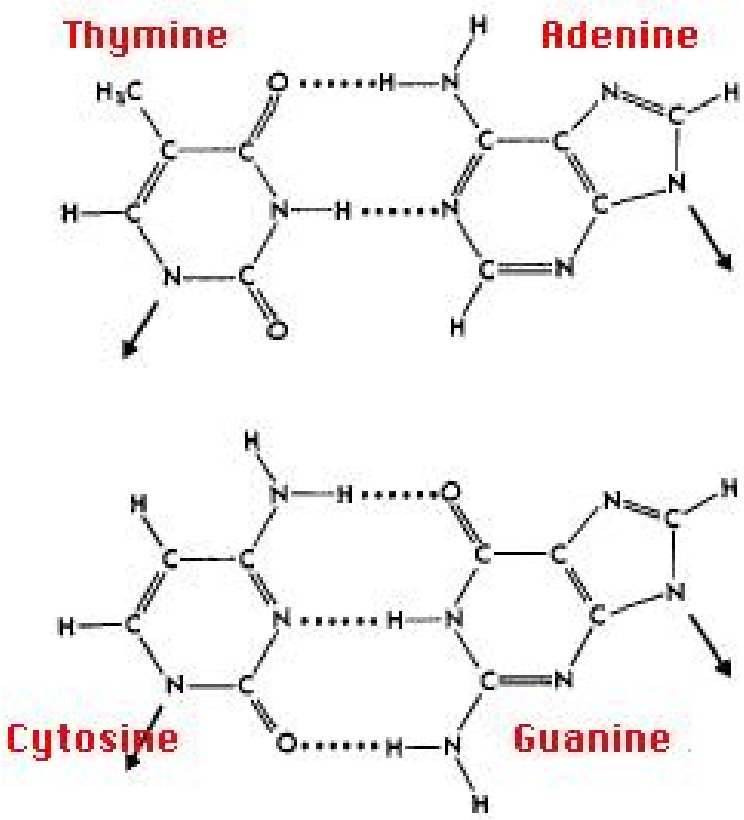
Regulation in Promoter Region

- Many different proteins can bind to the promoter region
- They can either excite the expression of the protein (by increasing the probability of RNA-polymerase binding)
- Or inhibit the expression of the protein

Binding Sites



TATA box

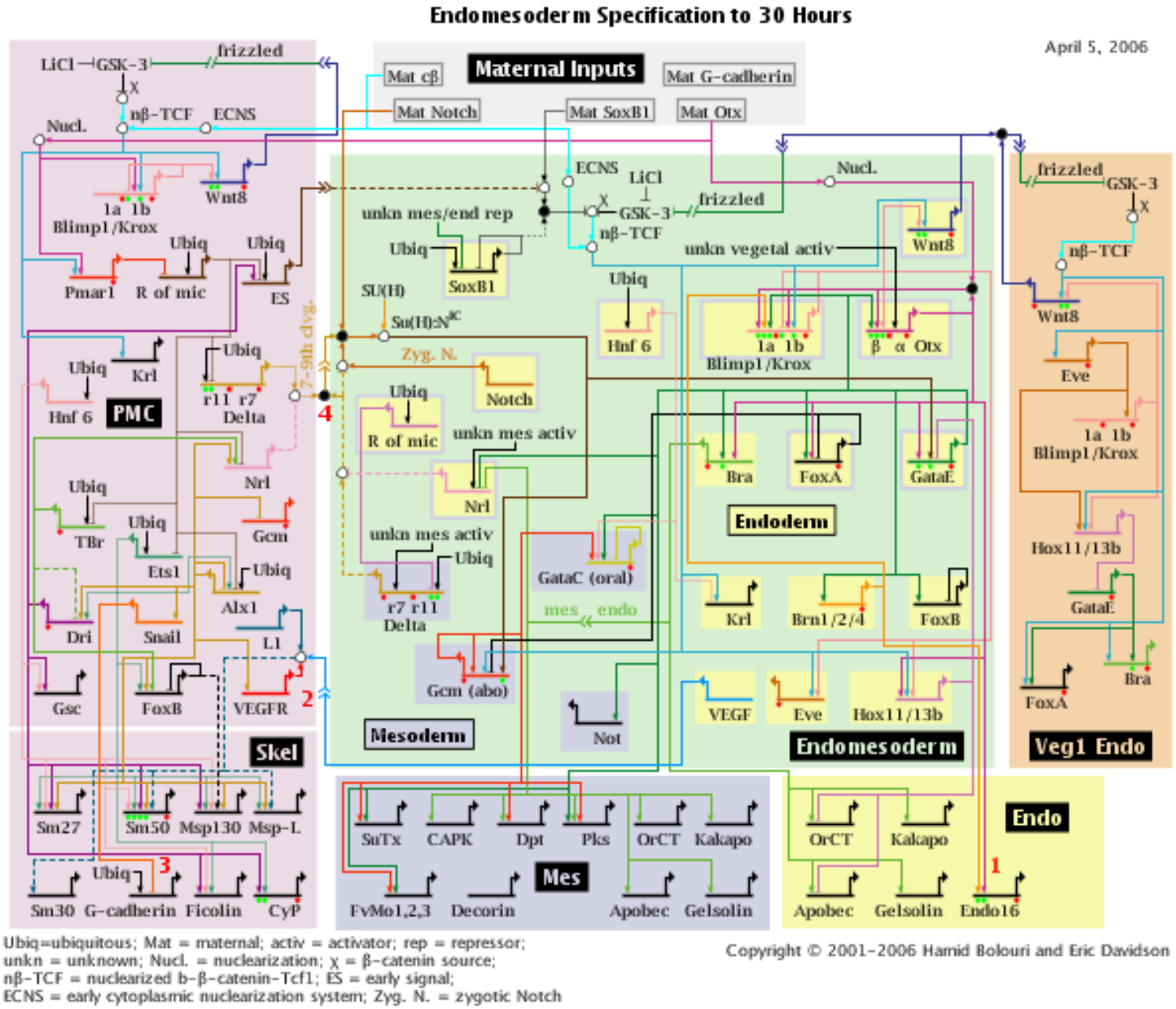


- Close to the coding regions is a region where there are more A-T pairs than usual
- Because A-T pairs have only 2 hydrogen bonds they are easier to break
- This helps RNA-polymerase split open the double helix

Gene Regulation Networks

- Proteins can regulate their own expression
- Proteins can regulate many other proteins
- Protein A can regulate protein B which in turn regulates protein A (if these are inhibitory then you get a flip-flop)
- In general the regulation forms a complex circuit

Sea Urchin Embryo GRN



Motifs

- To understand what is going on we can look for patterns or **motifs**
- We can spot common motifs by asking what patterns happen more than by chance
- Synthetic biology actually build biological circuits from these motifs
- We will study *gene regulation networks* (GRN) later in the course

Dynamics

- Gene regulation networks are dynamic
- To understand what is happening we need to simulate the network
- This requires knowledge of the rate at which the molecules are being produced
- We will explore the techniques for this later in the course too

Multicellular

- Many Eukaryotes are multicellular
- In many cases the cells differentiate
- How do cells know what role they should play?
- How do organisms develop so the cells do the right thing?
- We will look (rather superficially) at some answers to these questions

Summary

- We are at the very beginning of understanding how cells work
- Much of our understanding comes from comparing different organisms
- We are beginning to model some of the basic interactions that make cells function
- This needs a host of computation tools which we will learn about later in this course